

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

January 2022

ALGEBRA (Ph.D. Version)

Instructions to the student

- Answer all six questions; each will be assigned a grade from 0 to 10.
 - Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - Keep scratch work on separate pages in the same booklet.
-

- Show that every group of order $99 = 3^2 \cdot 11$ is abelian.
 - Describe, in terms of generators and relations, a non-abelian group of order $363 = 3 \cdot 11^2$. (*Hint:* The matrix $M = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$ satisfies $M^3 = I$.)
- Let V be an n -dimensional vector space over the complex numbers and let $T : V \rightarrow V$ be a linear transformation. Let $0 < m < n$.
 - Show that there exists a subspace S of V of dimension m such that $T(S) \subseteq S$.
 - Suppose $T(W) \subseteq W$ for all subspaces of dimension m . Show that T is a scalar multiple of the identity. (*Hint:* Every one-dimensional subspace can be written as the intersection of a set of m -dimensional subspaces (you do not need to prove this fact).)
- Let R be a commutative ring with 1. Suppose there is an exact sequence of R -modules
$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0.$$
Let $r, s \in R$ be such that the ideal generated by $\{r, s\}$ is R . Suppose that $rA = sC = 0$.
 - Show that the map $C \rightarrow C$ given by $c \mapsto rc$ is an isomorphism.
 - Show that the restriction of g to rB gives an isomorphism $rB \simeq C$.
 - Show that $B \simeq A \oplus C$.
- Let R be a Noetherian integral domain. An element of R is called *irreducible* if it cannot be factored as ab , where $a, b \in R$ and neither a nor b is a unit. Show that every nonzero element of R can be factored (not necessarily uniquely) as a product of irreducibles.

CONTINUED ON NEXT PAGE

5. Let p be a prime and let K be a field of characteristic 0. Suppose every polynomial in $K[x]$ whose degree is not a multiple of p has a root in K .
- (a) Show that the degree of every finite extension L/K is a power of p .
- (b) Suppose, in addition, that every polynomial in $K[x]$ of degree p has a root in K . Show that K is algebraically closed.
6. Let G be a finite group and let H be a subgroup of G . Let N be the smallest normal subgroup of G that contains H .
- (a) Let ρ be a finite-dimensional representation of G . Show that if H is in the kernel of ρ , then N is in the kernel of ρ .
- (b) Let

$$S = \{\rho \mid H \subseteq \text{Ker}(\rho)\},$$

where ρ runs through the irreducible representations of G . Let $\chi_\rho = \text{Trace}(\rho)$ be the character of ρ . Show that

$$[G : H] \geq \sum_{\rho \in S} \chi_\rho(1)^2,$$

with equality if and only if H is a normal subgroup of G .